

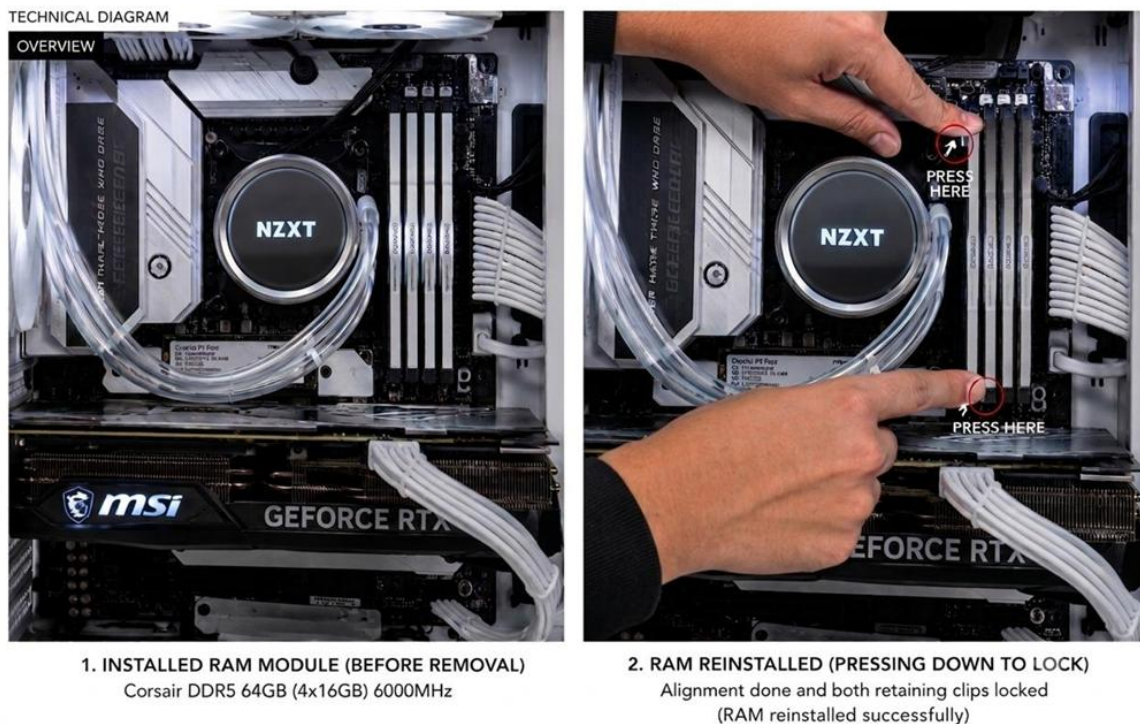
1.

Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3 RAM	DDR4 RAM	DDR5 RAM
Speed (Data Rate)	800 to 2,133 MT/s	1,600 to 3,200 MT/s	4,800 to 8,400+ MT/s
Voltage (Power)	1.5 Volts <i>(Highest power use)</i>	1.2 Volts <i>(Moderate power use)</i>	1.1 Volts <i>(Lowest power use)</i>
Typical Use Cases	Basic web browsing, light office work, and legacy/older desktop PCs	Standard everyday computing, 1080p gaming, office multitasking, and budget PC builds	4K/8K video editing, heavy 3D rendering, high-end gaming, AI, and server workloads

2.

Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.



1. Reinstall the RAM

- **Align the Notch:** Hold the RAM by its edges. Match the small gap (notch) on the golden connector of the RAM with the ridge in the slot.
- **Insert at an Angle:** Slide the RAM into the slot at a **45-degree angle** until it sits firmly in place.
- **Push Down to Lock:** Gently press down on the top edge of the RAM module until the side clips **click** automatically into place.

2. Reassemble the Device

- Put the back cover or side panel back on your computer.

- Tighten the screws and reconnect the power cable (or insert the battery).

3. Verify System Boot

- **Turn On:** Press the power button to turn on your system.
- **Check Display:** If your computer boots up normally to the desktop, the RAM was installed correctly.
- **Check Memory:**
 - Open **Task Manager** (press Ctrl + Shift + Esc).
 - Click on the **Performance** tab and select **Memory** to make sure your computer recognizes the full amount of installed RAM.

3.

List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

1. Mechanism & Speed:

- **HDD (Hard Disk Drive):** Uses spinning magnetic platters and a moving mechanical head. It is the slowest storage type (~100–150 MB/s).
- **SATA SSD (Solid State Drive):** Uses flash memory chips with no moving parts. It is moderately fast (~500–550 MB/s).
- **NVMe SSD (Non-Volatile Memory Express):** Uses flash memory attached directly to the high-speed PCIe bus on the motherboard. It is extremely fast (~3,500–7,000+ MB/s).

2. Latency & Response Time:

- **HDD:** High latency because the drive physically has to spin and move the head to find files.
- **SATA SSD:** Low latency with instantaneous access to data files.
- **NVMe SSD:** Ultra-low latency designed to process thousands of data queues simultaneously without bottlenecks.

3. Cost per Gigabyte:

- **HDD:** Cheapest option for bulk storage capacity.
- **SATA SSD:** Moderately priced balance between speed and budget.

- **NVMe SSD:** Higher cost per gigabyte, premium performance.

1. HDD — The Basement Pantry

- **How it works:** It's like walking down two flights of stairs to the basement pantry to grab an old jar of sauce. It takes time, but it holds tons of food for very little money.
- **Daily App: Storing old photos & movies (File Explorer / Downloads).** You only open these once in a while, so waiting a few seconds doesn't bother you.

2. SATA SSD — The Kitchen Cupboard

- **How it works:** It's like opening the cabinet right above your counter. You reach up and grab what you need in one second.
- **Daily App: Opening Google Chrome or Word.** Your computer turns on in seconds and web browsers open almost instantly.

3. NVMe SSD — The Countertop Blender

- **How it works:** It's right in front of you on the counter, already plugged in and running at full power. Zero waiting time.
- **Daily App: Heavy Gaming or Video Editing (Instagram Reels / YouTube creators).** Giant game maps load immediately without any frozen loading screens.

4.

Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

